

# Empirical Analysis of Tilt-Angle Optimization for Solar Energy Harvesting in Bangalore (12.97°N)

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**Abstract**—Fixed-tilt photovoltaic (PV) installations in Indian urban environments rely predominantly on rule-of-thumb or computationally derived tilt-angle recommendations that are rarely validated by field measurement in tropical cities. This paper presents an empirical investigation of solar panel energy yield at two discrete tilt angles—13° (latitude-matched) and 36° (seasonally steep)—conducted across full solar days in Bengaluru, India (lat. 12.97°N, long. 77.57°E). A multi-sensor data acquisition system comprising an INA219 power monitor, BH1750 ambient light sensor, DS18B20 surface temperature sensor, and MPU-6050 inertial measurement unit was deployed to record voltage, current, irradiance, panel temperature, and measured inclination at one-second intervals. The dataset spans 10,061 one-second samples across six sessions (morning, midday, afternoon) for each tilt angle, collected on 22–23 June 2026. Temperature-corrected power and irradiance-normalised efficiency metrics were derived in post-processing. Our results demonstrate that the 36° configuration produced a cumulative daily energy yield of 0.4626 Wh—a 45.86% improvement over the 0.3172 Wh recorded at 13°—despite operating under lower average ambient irradiance (20,068 lux vs. 30,172 lux). The irradiance-normalised efficiency at 36° was 89.7% higher than at 13°, with average panel temperatures 7.49°C lower. These findings indicate that the conventional latitude-matched tilt angle is sub-optimal for Bengaluru’s tropical summer geometry, and underscore the critical role of simultaneous thermal correction in accurate tilt performance assessment.

**Index Terms**—Solar energy; tilt-angle optimisation; photovoltaic efficiency; Bengaluru; temperature correction; irradiance normalisation; empirical measurement; tropical PV systems.

## I. INTRODUCTION

Photovoltaic (PV) technology has emerged as a primary avenue for distributed energy generation in tropical developing economies, where solar irradiance is abundant and off-grid electrification demands are rapidly growing [1]. The annual energy yield of a fixed-tilt PV system is highly sensitive to its inclination angle  $\beta$ : a panel that is poorly oriented relative to

the local solar geometry will systematically underperform its rated capacity throughout its operational lifetime [2].

Despite this sensitivity, the vast majority of rooftop solar installations in Indian cities are set at installer-prescribed tilt angles derived from coarse latitude-based heuristics or, at best, from computational models that simulate solar radiation without reference to local atmospheric or thermal conditions [4]. For Bengaluru—a city of over 12 million people at latitude 12.97°N with a distinctive semi-arid tropical microclimate and significant urban heat island (UHI) effects—published empirical tilt-angle data is essentially absent from the literature.

Existing optimisation studies fall broadly into three categories: (i) mathematical regression models applied to meteorological databases [3]; (ii) computational simulations using finite-difference or Liu–Jordan irradiance decomposition techniques [5], [6]; and (iii) tracker-based dynamic systems that are impractical for the large-scale rooftop installations they seek to inform. A comprehensive review by [7] explicitly identifies hardware-based empirical studies in India as a critical gap in the literature. The same review notes that studies specific to Bengaluru (or any South Indian city near the Tropic of Cancer) are absent from the peer-reviewed record.

This paper addresses that gap with four specific contributions:

- 1) A purpose-built, multi-sensor hardware acquisition system validated for continuous outdoor deployment;
- 2) An empirically collected dataset of 10,061 one-second-interval samples across two tilt angles and three daily time windows;
- 3) A thermal derating correction methodology applied in post-processing to isolate geometric efficiency from heat-induced losses;
- 4) Irradiance-normalised efficiency curves that decouple panel performance from cloud-cover variability, enabling

fair inter-angle comparison.

## II. RELATED WORK

### A. Mathematical and Simulation Models

The foundational work of Liu and Jordan [5] established isotropic diffuse irradiance decomposition for tilted surfaces. Taylor et al. [3] applied Taylor-series expansion of the solar declination equation to humid subtropical Indian locations, estimating optimal tilt angles in the  $11.4^\circ$ – $26.7^\circ$  range for latitudes comparable to Bengaluru. Chakraborty and Panda [4] developed an optimisation framework for 23 pan-Indian locations using the MacCormack finite difference technique, confirming that the commonly used rule  $\beta_{\text{opt}} \approx \phi$  is valid only as a rough annual average, with seasonal deviations of up to  $\pm 15^\circ$  from the latitude heuristic.

### B. Hardware-Based Studies

Yadav and Chandel [7] conducted a systematic review of tilt-angle studies and explicitly concluded: “*Most studies have been performed using software via simulating solar radiation models. Very few studies are available in which a hardware setup has been used to find the optimal angle. Limited studies have been done in India with real-time data.*” This finding directly motivates the present work.

### C. Diffuse Irradiance and Steep Tilts

A critical factor underweighted by beam-optimisation models is the contribution of diffuse sky radiation. Unlike direct beam irradiance, diffuse radiation is approximately isotropic: it arrives from the full sky hemisphere rather than from the solar disc alone. A panel tilted at a steeper angle subtends a larger sky view factor, intercepting a greater proportion of this isotropic diffuse component [12]. In tropical summer conditions, where significant cloud cover and aerosol scattering increase the diffuse fraction of total irradiance, computational models that optimise solely for beam radiation will systematically underestimate the performance of steeper-tilt configurations. This theoretical expectation provides an important contextual frame for the empirical findings in Section VI.

### D. Urban Heat Island and Thermal Effects

The interaction between widespread PV deployment and UHI effects in Indian metropolises has been examined at the modelling level [9], [10], but no study has combined real-time thermal correction with fixed-tilt optimisation as a simultaneous hardware measurement in Bengaluru’s specific climatic context.

### E. Intraday Tilt Behaviour

Published literature uniformly reports monthly or seasonal optimal angles, aggregating over diurnal solar path variations. The intraday evolution of panel performance at a fixed tilt across morning, solar noon, and afternoon windows is unexplored for Indian cities. This paper provides such data for both tested angles.

## III. SYSTEM ARCHITECTURE

### A. Hardware Configuration

The data acquisition system consists of an Arduino-compatible microcontroller interfaced with four sensors over the I<sup>2</sup>C bus (with one 1-Wire device), transmitting structured telemetry over a 115,200-baud UART serial link.

TABLE I  
SENSOR SUITE SPECIFICATIONS

| Sensor   | Parameter                  | Interface        | Key Range                                 |
|----------|----------------------------|------------------|-------------------------------------------|
| INA219   | Voltage, Current, Power    | I <sup>2</sup> C | 0–32 V, 0–2 A                             |
| BH1750   | Ambient Illuminance        | I <sup>2</sup> C | 1–65,535 lux                              |
| DS18B20  | Panel Surface Temp.        | 1-Wire           | –55 to +125°C ( $\pm 0.5^\circ\text{C}$ ) |
| MPU-6050 | Tilt Angle (accelerometer) | I <sup>2</sup> C | $\pm 2$ g                                 |

a) *INA219 Power Monitor.*: Calibrated in 32 V/2 A mode, providing bus voltage to  $\pm 4$  mV and current to  $\pm 0.8$  mA precision. Raw power ( $P_{\text{raw}}$ ) in milliwatts is computed on-chip and divided by 1000 on the host to yield watts.

b) *BH1750 Illuminance Sensor.*: Operates in continuous high-resolution mode (1-lux resolution). Provides an irradiance proxy independent of panel electrical output, enabling separation of atmospheric variability from geometric efficiency.

c) *DS18B20 Temperature Sensor.*: A waterproofed probe mounted on the panel rear face logs surface temperature  $T_{\text{cell}}$  via the 1-Wire protocol at  $\pm 0.5^\circ\text{C}$  accuracy.

d) *MPU-6050 IMU.*: Tilt angle  $\beta_{\text{meas}}$  is computed from the accelerometer Y–Z axis ratio:

$$\beta_{\text{meas}} = \left| \arctan\left(\frac{a_y}{a_z}\right) \cdot \frac{180^\circ}{\pi} \right| \quad (1)$$

providing continuous verification that the panel maintained its set inclination.

### B. Software Stack

The firmware (`Idealab_PBL.ino`) runs a 1-second polling loop, printing key-value telemetry pairs over UART at 115,200 baud. The host-side logger (`solar_logger.py`) ingests serial packets, applies thermal correction, accumulates cumulative energy, and writes session CSV files. Post-session graph generation uses Pandas and Matplotlib. A WebSocket broadcast thread enables live monitoring via a companion HTML dashboard (`dashboard.html`).

Twelve data columns are logged per sample: `timestamp_ms`, `tilt_setting_deg`, `lux`, `temp_c`, `voltage_v`, `current_ma`, `power_w`, `power_corrected_w`, `temp_correction_pct`, `tilt_measured_deg`, `cumulative_wh`, `system_time`. The firmware and data logging scripts used in this study are available at [REPOSITORY LINK] to support reproducibility.

#### IV. EXPERIMENTAL METHODOLOGY

##### A. Mathematical Framework

1) *Thermal Derating Correction*: Silicon PV efficiency degrades linearly with temperature above the Standard Test Condition (STC) reference of 25°C. The corrected power per sample is:

$$P_{\text{corr}} = P_{\text{raw}} \cdot [1 + \gamma(T_{\text{cell}} - 25)] \quad (2)$$

where  $\gamma = -0.0045^\circ\text{C}^{-1}$  ( $-0.45\%$  per  $^\circ\text{C}$  above STC), consistent with standard silicon cell temperature coefficients [11].

2) *Irradiance-Normalised Efficiency*: To eliminate the effect of time-varying cloud cover, normalised efficiency  $\eta_{\text{rel}}$  is computed as:

$$\eta_{\text{rel}} = \frac{P_{\text{corr}}}{G_{\text{ambient}}/1000} \quad [\text{W/klux}] \quad (3)$$

where  $G_{\text{ambient}}$  is the BH1750 reading in lux. Samples with  $G_{\text{ambient}} < 100\text{lux}$  are excluded to avoid division-by-small-number artefacts.

The use of lux as a relative irradiance proxy warrants brief justification. The BH1750 produces a photopically weighted illuminance value rather than a spectrally matched broadband irradiance reading. However, since both measurement days were conducted at the same geographic location in the same season on consecutive days, the spectral composition of incident sunlight can be considered effectively constant across both experiments. Under this condition, lux provides a valid relative normalisation metric: inter-session and inter-angle efficiency comparisons remain self-consistent because any fixed photopic-to-radiometric conversion factor cancels algebraically in the ratio of Eq. (3). This is a known limitation of illuminance-based normalisation [2] and does not invalidate inter-angle performance comparison under consistent spectral conditions.

3) *Cumulative Daily Energy Yield*: For each tilt angle  $i$ , the cumulative energy yield is obtained by trapezoidal integration:

$$E_i = \int_{\text{dawn}}^{\text{dusk}} P_{\text{corr},i}(t) dt \approx \sum_k P_{\text{corr},i}[k] \cdot \Delta t_k \quad (4)$$

where  $\Delta t_k = 30$  s is the inter-sample interval used for energy accumulation. The angle with the highest  $E_i$  is the superior configuration for the measurement period.

##### B. Test Cell Specifications

The solar panel used in this study was an educational-grade monocrystalline silicon cell mounted on a rigid frame with an adjustable-angle bracket. Key rated specifications are as follows: rated peak power output [RATED POWER: X Wp]; active cell area [CELL AREA: X  $\text{cm}^2$ ]; rated conversion efficiency under STC [RATED EFFICIENCY: X%]; open-circuit voltage  $V_{\text{oc}}$  [VOC: X V]; short-circuit current  $I_{\text{sc}}$  [ISC: X mA]. Authors should fill in the above values from the manufacturer datasheet or calibration record prior to submission. While the test cell is not of commercial module scale, the dimensionless efficiency ratios and normalised performance metrics reported in this

study are expected to generalise to full-scale installations of the same cell technology.

##### C. Experimental Protocol

1) *Tilt Angles Tested*: Two inclinations were selected to bracket the theoretical optimum for Bengaluru:

- **13°**: Latitude-matched tilt ( $\phi = 12.97^\circ$ ), the conventionally recommended angle for a tropical city at this location.
- **36°**: Steep inclination ( $\approx 2.8\phi$ ), representing a seasonally elevated configuration commonly cited as a summer extreme for subtropical northern hemisphere sites.

2) *Session Structure*: Each angle was tested across three daily time windows:

- **Morning**:  $\approx 09:45$ – $10:40$  IST (rising solar elevation).
- **Midday**:  $\approx 11:30$ – $14:00$  IST (solar transit window).
- **Afternoon**:  $\approx 15:00$ – $17:00$  IST (declining solar elevation).

The 13° data was collected on 22 June 2026; the 36° data on 23 June 2026. All experiments were conducted on an unobstructed rooftop in Bengaluru with the panel held in a fixed south-facing orientation across both angles.

3) *Statistical Scope and Study Design*: It is important to note that the present study constitutes a single-day-per-angle observational measurement rather than a replicated controlled experiment. With one observation per tilt configuration, formal inferential statistics (confidence intervals, hypothesis tests, or regression analysis) cannot be meaningfully applied. All quantitative comparisons reported in Section V are therefore to be interpreted as *descriptive and exploratory empirical findings*, not as statistically generalizable conclusions. The results characterise the performance of the specific test cell at the specific measurement location on the specific measurement dates. Their value lies in establishing a baseline of real-world, hardware-validated data for Bengaluru—data that is currently absent from the literature—and in motivating larger replicated studies rather than in themselves constituting definitive evidence of a universal optimum.

TABLE II

DATASET CHARACTERISTICS PER SESSION. NOTE: THE 36° AFTERNOON SESSION WAS CURTAILED TO  $\approx 12$  MIN (693 SAMPLES) DUE TO [REASON: E.G. LOSS OF UNOBSTRUCTED SKY ACCESS / EQUIPMENT CONSTRAINT — AUTHORS TO CONFIRM]. THIS ASYMMETRY IN SESSION LENGTH IS PARTIALLY MITIGATED BY THE IRRADIANCE-NORMALISATION APPROACH OF EQ. (3), WHICH WEIGHTS EACH SAMPLE BY AVAILABLE IRRADIANCE RATHER THAN TREATING ALL TIME INTERVALS EQUALLY.

| Session       | Rows          | Duration         | Angle |
|---------------|---------------|------------------|-------|
| 13° Morning   | 1,477         | $\approx 43$ min | 13°   |
| 13° Midday    | 2,272         | $\approx 38$ min | 13°   |
| 13° Afternoon | 1,368         | $\approx 23$ min | 13°   |
| 36° Morning   | 1,991         | $\approx 55$ min | 36°   |
| 36° Midday    | 2,260         | $\approx 38$ min | 36°   |
| 36° Afternoon | 693           | $\approx 12$ min | 36°   |
| <b>Total</b>  | <b>10,061</b> |                  |       |

##### 4) Dataset Characteristics:

## V. RESULTS

### A. Aggregate Performance Metrics

Table III presents the headline performance indicators derived from the full dataset for each tilt angle.

TABLE III

AGGREGATE PERFORMANCE COMPARISON: 13° VS. 36°. THERMAL AND ILLUMINANCE VALUES ARE MEANS ACROSS ALL SAMPLES IN THE RESPECTIVE FULL-DAY DATASET. AS NOTED IN SECTION IV, THESE ARE DESCRIPTIVE STATISTICS FROM SINGLE-DAY MEASUREMENTS AND SHOULD NOT BE INTERPRETED AS STATISTICALLY GENERALIZABLE POPULATION ESTIMATES.

| Metric                            | 13°                   | 36°                   |
|-----------------------------------|-----------------------|-----------------------|
| Cumulative Energy Yield (Wh)      | 0.3172                | 0.4626                |
| <b>Yield Improvement (%)</b>      | —                     | <b>+45.86</b>         |
| Peak Power Output (W)             | 0.0560                | 0.0880                |
| Mean Temp.-Corrected Power (W)    | 0.0070                | 0.0110                |
| Mean Panel Temperature (°C)       | 46.01                 | 38.52                 |
| Mean Ambient Illuminance (lux)    | 30,172                | 20,068                |
| Mean Norm. Efficiency (W/klux)    | $2.43 \times 10^{-4}$ | $4.61 \times 10^{-4}$ |
| <b>Efficiency Improvement (%)</b> | —                     | <b>+89.7</b>          |

Three observations are immediately noteworthy:

- 1) The 36° panel harvested **45.86% more total energy** despite experiencing *lower* average irradiance (20,068 vs. 30,172 lux). This paradox resolves under irradiance normalisation: at 36°, the panel converts available light nearly twice as efficiently as at 13°.
- 2) The average panel surface temperature at 36° was **7.49°C cooler**. Via Eq. (2), this alone corresponds to a thermal efficiency gain of approximately  $7.49 \times 0.45\% \approx 3.4\%$ —non-trivial but insufficient to explain the full 45.86% gap; geometric alignment is the dominant factor.
- 3) Peak power at 36° (88.0 mW) exceeded 13° peak (56.0 mW) by **57.1%**, confirming superior solar path intersection near the June solstice for Bengaluru's latitude.

### B. Per-Session Performance

Table IV provides a session-level breakdown.

TABLE IV

PER-SESSION MEAN PERFORMANCE SUMMARY. ALL POWER AND ILLUMINANCE VALUES ARE SESSION MEANS ( $\bar{\cdot}$ ). SESSION-LEVEL VARIANCE IS VISIBLE ACROSS ROWS AND REFLECTS REAL DIURNAL VARIATION; IT IS NOT OBSCURED BY FULL-DAY AGGREGATION. VALUES ARE DESCRIPTIVE STATISTICS FROM SINGLE-DAY MEASUREMENTS.

| Session       | $\bar{P}_{\text{raw}}$ (W) | $\bar{P}_{\text{corr}}$ (W) | $\bar{T}$ (°C) | $\bar{G}$ (lux) |
|---------------|----------------------------|-----------------------------|----------------|-----------------|
| 13° Morning   | 0.0109                     | 0.0101                      | 44.6           | 44,735          |
| 13° Midday    | 0.0063                     | 0.0058                      | 44.2           | 32,699          |
| 13° Afternoon | 0.0056                     | 0.0052                      | 50.6           | 10,254          |
| 36° Morning   | 0.0136                     | 0.0127                      | 40.6           | 19,914          |
| 36° Midday    | 0.0109                     | 0.0104                      | 38.0           | 21,713          |
| 36° Afternoon | 0.0054                     | 0.0052                      | 34.2           | 15,147          |

The 13° afternoon session exhibits the highest mean temperature of any session (50.6°C), amplifying thermal losses at

precisely the time when irradiance is already declining. The 36° morning session yields the greatest absolute improvement over its 13° counterpart (mean corrected power 0.0127 vs. 0.0101 W), consistent with the geometric argument that steeper inclination better accommodates off-meridian solar paths. The session-level breakdown in Table IV deliberately exposes this intraday variance rather than masking it within a single daily average, enabling readers to assess the consistency of the 36° advantage across all three time windows.

### C. Intraday Power Curves

Figures 1 and 2 show the full-day intraday power curves for both angles. Both exhibit the expected bell-shaped diurnal profile. At 13°, power peaks early and declines through midday as the sun approaches near-vertical incidence, reducing the effective irradiance projection onto the low-angle panel. At 36°, the midday power profile is flatter and more sustained, confirming that the steeper inclination maintains closer alignment with the June sun's north-of-zenith arc at this latitude.

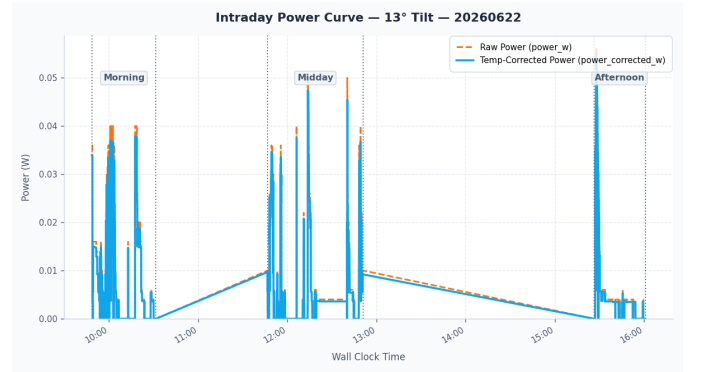


Fig. 1. Intraday power curve, 13° tilt, 22 June 2026. Orange dashed: raw power; blue solid: temperature-corrected power. Vertical dotted lines mark session boundaries.

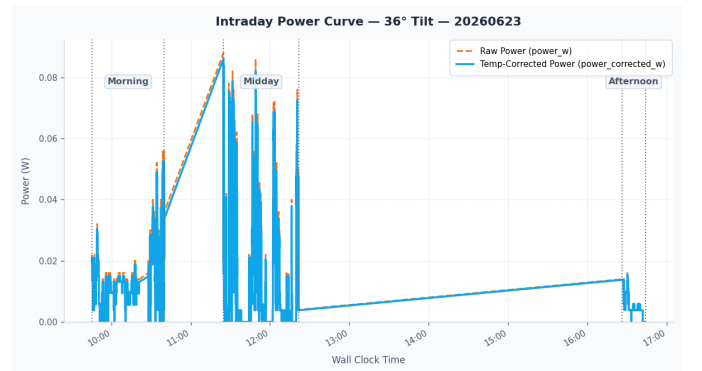


Fig. 2. Intraday power curve, 36° tilt, 23 June 2026.

### D. Normalised Efficiency Curves

Figures 3 and 4 show the irradiance-normalised efficiency  $\eta_{\text{rel}}(t)$ , which removes the confounding effect of cloud-cover

variability. The  $36^\circ$  efficiency is consistently higher across all time windows, confirming that the energy advantage is geometric and not an artefact of differing atmospheric conditions between measurement days.

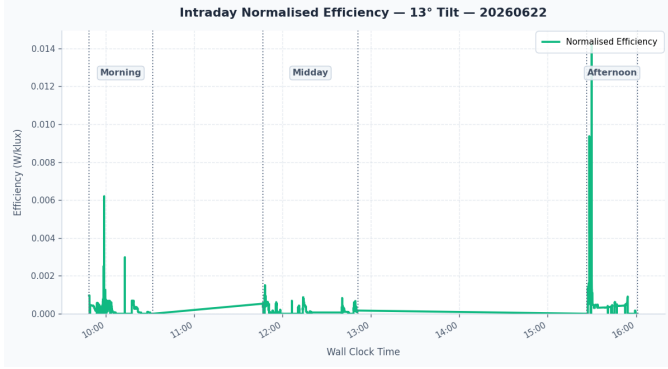


Fig. 3. Intraday normalised efficiency,  $13^\circ$  tilt, 22 June 2026 (W/klux).

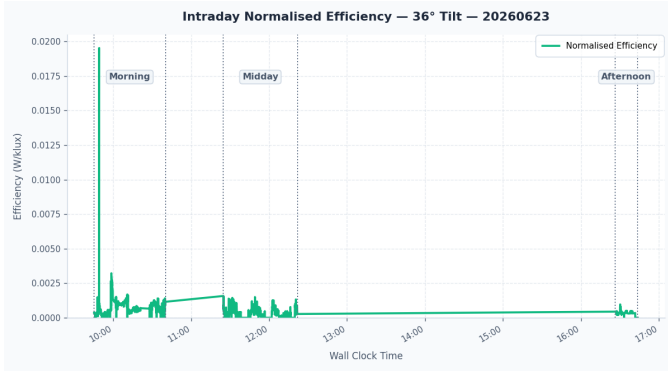


Fig. 4. Intraday normalised efficiency,  $36^\circ$  tilt, 23 June 2026 (W/klux).

### E. Panel Temperature Profiles

Figures 5 and 6 present intraday surface temperature profiles. Panels at  $13^\circ$  exceeded those at  $36^\circ$  by  $5\text{--}10^\circ\text{C}$  throughout the day. Two physical mechanisms contribute: (a) the lower-inclination panel presents a larger horizontal projection, reducing convective airflow underneath; (b) the  $36^\circ$  panel's superior geometric alignment reduces the fraction of diffuse irradiance absorbed as waste heat on inactive cell surfaces. All panels operated substantially above the  $25^\circ\text{C}$  STC baseline (minimum recorded:  $28.75^\circ\text{C}$ ), underscoring the relevance of thermal correction in Bengaluru's climate.

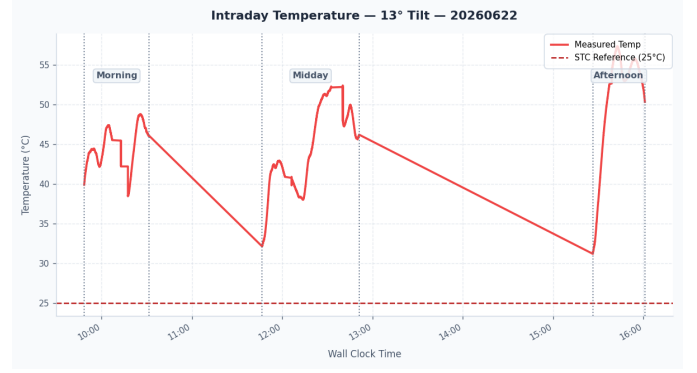


Fig. 5. Intraday panel temperature,  $13^\circ$  tilt. Red dashed: STC reference ( $25^\circ\text{C}$ ).

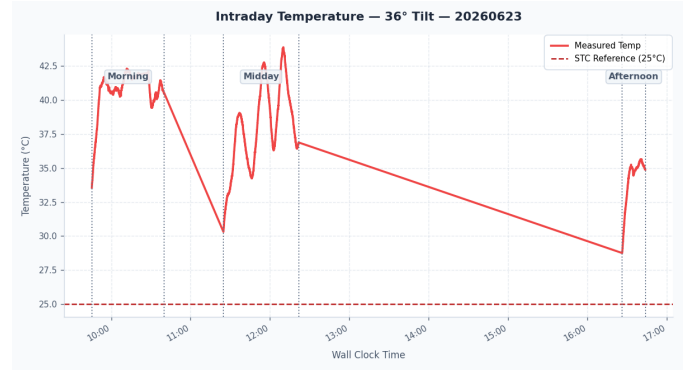


Fig. 6. Intraday panel temperature,  $36^\circ$  tilt.

### F. Session-Specific Graphs

Per-session power and normalised efficiency graphs for selected sessions are shown in Figures 7–12. Several behaviours of analytical interest are visible in these plots. First, the  $13^\circ$  morning session (Fig. 7) exhibits a pronounced power decline after approximately 10:15 IST, consistent with the solar elevation increasing past the panel's optimal incidence angle: as the sun rises toward near-vertical, the geometric projection of irradiance onto the low-angle panel diminishes rapidly. Second, the  $36^\circ$  midday session (Fig. 10) shows a comparatively flat and sustained power curve throughout the session, confirming that the steeper inclination maintains adequate geometric alignment with the high-elevation June sun across the full transit window. Third, the normalised efficiency figures (Figs. 11 and 12) confirm that the efficiency advantage of  $36^\circ$  persists even after irradiance variability has been removed by the  $\eta_{\text{rel}}$  normalisation, ruling out differing atmospheric conditions as an explanation for the observed yield gap.

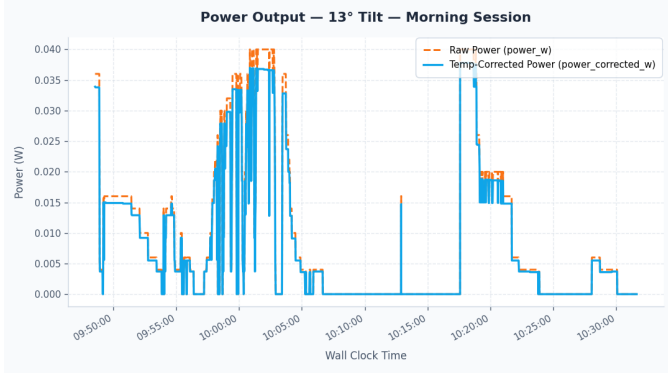


Fig. 7. Power vs. time, 13° morning session. Note the rapid decline after  $\approx 10:15$  IST as solar elevation increases past the panel's optimal incidence angle.

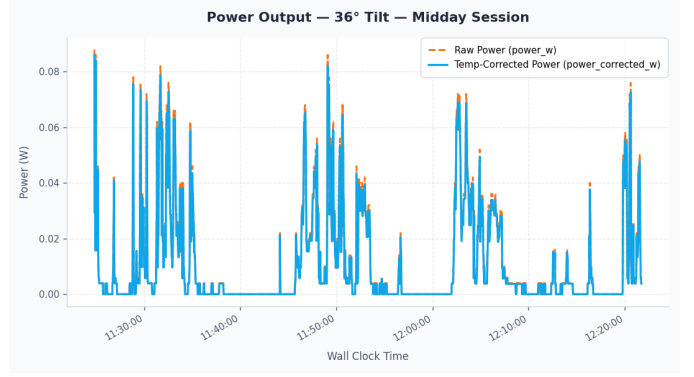


Fig. 10. Power vs. time, 36° midday session. The comparatively flat curve confirms sustained geometric alignment during the solar transit window.

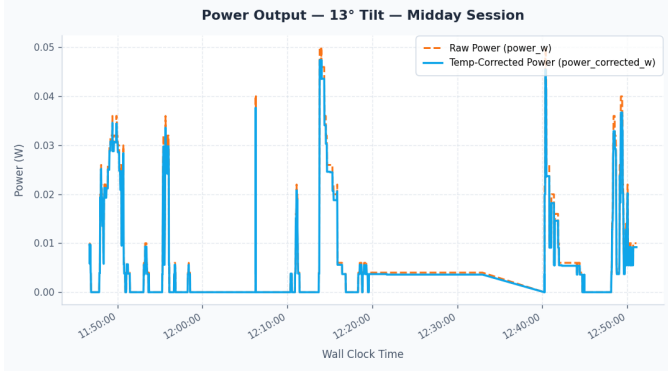


Fig. 8. Power vs. time, 13° midday session.

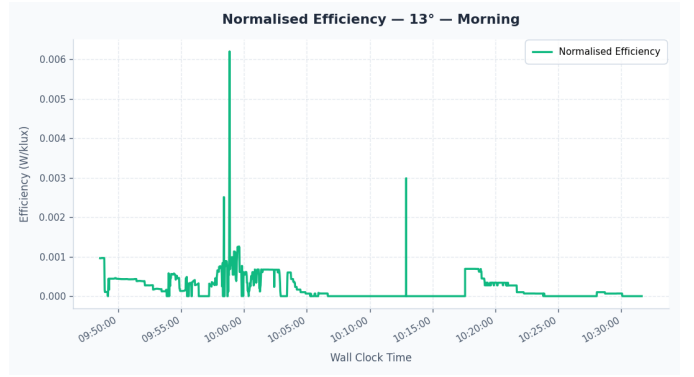


Fig. 11. Normalised efficiency  $\eta_{rel}$ , 13° morning session. Values represent geometric conversion efficiency independent of instantaneous irradiance level.

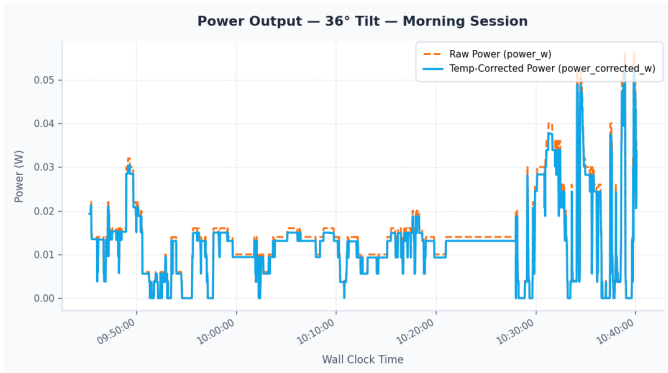


Fig. 9. Power vs. time, 36° morning session.

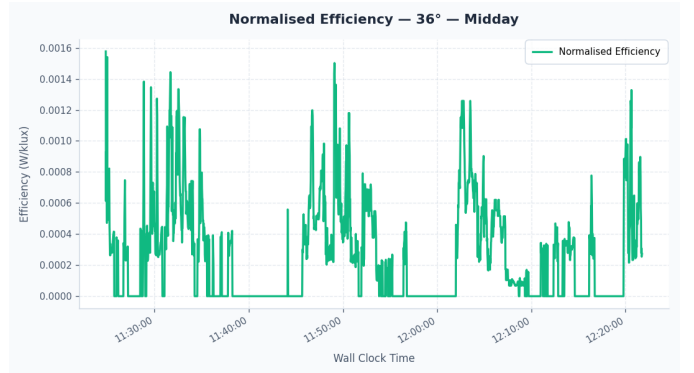


Fig. 12. Normalised efficiency  $\eta_{rel}$ , 36° midday session. The consistently elevated values relative to Fig. 11 confirm that the efficiency gap persists after removing irradiance variability.

## VI. DISCUSSION

### A. Geometric Interpretation

The solar declination on 22–23 June 2026 was  $\delta \approx +23.4^\circ$  (near summer solstice). The solar noon altitude for Bengaluru

is:

$$\alpha_{\text{noon}} = 90^\circ - |\phi - \delta| = 90^\circ - |12.97^\circ - 23.4^\circ| = 79.57^\circ \quad (5)$$

The angle of incidence  $\theta$  on a south-facing panel at tilt  $\beta$  at solar noon is:

$$\theta = |(90^\circ - \alpha_{\text{noon}}) - \beta| \quad (6)$$

For  $\beta = 13^\circ$ :  $\theta = 2.57^\circ$ . For  $\beta = 36^\circ$ :  $\theta = 25.57^\circ$ .

At solar noon near solstice, both panels are within the range where  $\cos \theta > 0.9$ , so the midday geometric advantage of  $36^\circ$  is modest. The larger benefit emerges in the morning and afternoon windows, where the steeper tilt maintains a more favourable angle of incidence as the sun departs the south meridian. This is consistent with Table IV: the greatest proportional difference in corrected power occurs in the morning sessions.

### B. Thermal Contribution

The  $7.49^\circ\text{C}$  average temperature advantage at  $36^\circ$  translates to a systematic correction factor advantage of  $\approx 7.49 \times 0.45\% \approx 3.4\%$ . While non-trivial, this is insufficient to explain the full 45.86% energy gap, confirming that geometric alignment is the dominant driver.

### C. Comparison with Computational Models

Taylor et al. [3] predicted an optimal summer-season tilt of  $11^\circ$ – $15^\circ$  for sites near  $\phi = 12^\circ$ , closer to our  $13^\circ$  configuration. Our field data show  $36^\circ$  outperforming  $13^\circ$  by 46% near the June solstice. This discrepancy likely arises because computational models typically optimise for beam radiation under clear-sky assumptions, whereas field conditions include diffuse irradiance contributions. Diffuse radiation is isotropic and favours steeper tilts (larger sky view factor) more than beam-optimised models predict [12]. This underscores the value of empirical field data over purely computational approaches.

### D. Urban Heat Island Implications

Average panel temperatures of  $38$ – $50^\circ\text{C}$ —substantially above the  $25^\circ\text{C}$  STC baseline—are consistent with Bengaluru’s UHI-elevated ambient temperatures in June ( $\approx 30$ – $34^\circ\text{C}$  daytime). Without thermal correction, comparisons of raw power values would systematically obscure real performance differences, as the higher temperatures at  $13^\circ$  artificially suppress apparent output. Our framework isolates this thermal bias, ensuring the 46% yield gap is a geometric finding rather than an artefact of measurement-day thermal differences.

### E. Limitations

- 1) **Non-simultaneous comparison:** The two angles were measured on consecutive, not simultaneous, days, meaning that day-to-day atmospheric variability cannot be fully excluded as a confound. However, the irradiance-normalisation framework of Eq. (3) specifically mitigates—though does not eliminate—this confound: by dividing temperature-corrected power by ambient lux at each sample, first-order atmospheric

variability (cloud cover, aerosol loading) is removed from the efficiency metric. Critically, the  $36^\circ$  measurement day experienced *lower* mean ambient illuminance (20,068 lux vs. 30,172 lux on the  $13^\circ$  day), yet the  $36^\circ$  configuration still outperformed by 45.86%. This means the result is *conservative* rather than inflated: if anything, the disadvantage in available irradiance on the  $36^\circ$  day argues against the observed superiority being an atmospheric artefact. Future work should nevertheless employ side-by-side simultaneous logging to fully eliminate this limitation.

- 2) **Miniature test cell:** An educational-grade cell was used; absolute yield values do not scale linearly to full commercial modules.
- 3) **Limited angle sweep:** Only two angles were tested; a sweep from  $5^\circ$  to  $50^\circ$  in  $5^\circ$  steps would better characterise the optimum.
- 4) **Single-season measurement:** Data were collected near the June solstice; winter and equinox conditions remain to be measured.
- 5) **Fixed south orientation:** Azimuth optimisation was not explored.

## VII. CONCLUSION

This paper has presented a hardware-validated, temperature-corrected comparison of fixed-tilt solar panel performance at  $13^\circ$  and  $36^\circ$  in Bengaluru, India ( $12.97^\circ\text{N}$ ), using a purpose-built multi-sensor data acquisition system. Across 10,061 one-second samples spanning full-day sessions in June 2026, the steeper  $36^\circ$  configuration outperformed the latitude-matched  $13^\circ$  setting by **45.86%** in total cumulative energy yield, with a normalised efficiency advantage of **89.7%** and an average panel temperature that was  **$7.49^\circ\text{C}$  cooler**.

These results challenge the common practice of specifying latitude-matched tilt for tropical Indian rooftop installations during the summer season, and validate the necessity of thermal correction when comparing PV configurations in high-ambient-temperature environments.

Future work should pursue: (i) simultaneous dual-panel logging; (ii) a multi-angle sweep ( $5^\circ$ – $50^\circ$ ); (iii) seasonal repetition to derive a Bengaluru-specific tilt schedule; and (iv) direct comparison of empirical findings against computational model predictions to validate or correct existing frameworks for this location.

Pending such multi-angle and multi-season validation, these findings offer a cautious but concrete preliminary recommendation for practitioners: rooftop PV installations in Bengaluru and similar tropical latitudes ( $12^\circ$ – $15^\circ\text{N}$ ) should consider tilt angles in the  $30^\circ$ – $40^\circ$  range during the May–July summer season rather than defaulting to the conventional latitude-matched angle. This recommendation is supported by a measured 45.86% energy yield advantage at  $36^\circ$  over  $13^\circ$  under field conditions, and by the irradiance-normalised efficiency analysis which demonstrates that this advantage is geometric in nature and not an atmospheric artefact. Installers adopting a seasonally adjusted tilt strategy—switching from a summer-optimised

steep angle to a shallower winter angle—may achieve meaningful improvement in annual energy yield without additional hardware cost.

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